

AI-Powered Burmese Voice Support Copilot for Banking

Intelligent AI Customer Support Solution

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A professional proposal for a secure Burmese AI support platform that converts approved banking knowledge into fast, consistent, and natural text-and-voice assistance.

1. Executive Summary

EXECUTIVE SUMMARY

A RAG-based Burmese banking copilot that delivers approved answers in text and voice, improving service consistency while reducing support effort and customer wait times.

The AI-Powered Burmese Voice Support Copilot is an intelligent customer-support solution designed for banks. Users enter questions as text and receive accurate responses in both Burmese text and natural Burmese speech.

The product uses Retrieval-Augmented Generation (RAG) to retrieve information from the bank's approved FAQs, policies, product documents, service procedures, and internal knowledge sources before generating a response. This improves accuracy, consistency, and trust compared with a general-purpose chatbot.

For Burmese semantic search, the solution uses a **Qwen3 Embedding model**. Document embeddings and searchable knowledge chunks are stored in **ChromaDB**, a vector database suitable for rapid prototyping, local deployment, and semantic retrieval.

2. Product Vision

To provide a secure, accurate, friendly, and natural Burmese-speaking banking assistant that improves customer experience while reducing the workload of customer-service teams.

3. Business Objectives

- Provide fast and consistent answers to common banking questions.
- Deliver answers in Burmese text and Burmese voice.
- Reduce customer waiting time.
- Reduce repetitive work for call-center and branch employees.
- Improve access to banking information for Burmese-speaking customers.
- Provide 24/7 support for approved use cases.
- Create a reusable AI platform that can later support voice input and banking API integrations.

4. Problems Addressed

Banks commonly face:

- High volumes of repetitive customer inquiries.
- Long customer waiting times.
- Inconsistent answers across agents and branches.
- Difficulty searching large policy and procedure documents.
- Limited high-quality Burmese-language support.
- Customers who prefer listening to instructions rather than reading long answers.
- Time-consuming onboarding for new customer-service staff.

The Support Copilot provides a single, searchable, voice-enabled knowledge interface for these challenges.

5. Target Users

External Users

- Mobile banking customers
- Internet banking customers
- Elderly customers
- Visually impaired customers
- Customers who prefer Burmese voice guidance

Internal Users

- Call-center agents
- Branch employees
- Customer-service representatives
- Sales representatives
- Relationship managers
- Operations and support teams

6. Core Product Features

6.1 Burmese Text Question Input

Users can ask questions naturally in Burmese or supported mixed Burmese-English text.

Example:

ATM ကတ်ပျက်သွားရင် ဘာလုပ်ရမလဲ။

6.2 Grounded Banking Answers

The system retrieves relevant information from approved bank documents before generating an answer.

Supported knowledge sources may include:

- Frequently asked questions
- Account-opening procedures
- Card replacement procedures
- Mobile banking guides
- Loan product information
- Service fees
- Customer-service procedures

- Internal standard operating procedures

6.3 Burmese Voice Response

Each supported answer can be returned as:

- Burmese text
- Burmese audio
- Referenced source information

The desired voice style is:

- Friendly
- Gentle
- Warm
- Calm
- Professional
- Conversational
- Easy to understand

6.4 Semantic Search with Qwen3 Embeddings

The Qwen3 Embedding model converts questions and document chunks into vector representations. This allows the system to retrieve information by meaning rather than relying only on exact keyword matches.

This is important for Burmese because customers may ask the same question using different wording.

6.5 ChromaDB Vector Storage

ChromaDB stores:

- Document chunks
- Qwen3-generated embeddings
- Source metadata
- Document names
- Page or section references
- Version information
- Access-control metadata where required

ChromaDB is suitable for:

- Local development
- Fast MVP delivery
- Docker-based deployment
- Persistent vector storage
- Metadata filtering
- Semantic similarity search

6.6 Conversation Context

The assistant maintains short-term conversation context so follow-up questions can be interpreted correctly.

Example:

1. “How do I replace my ATM card?”
2. “How much does it cost?”

The system understands that the second question concerns ATM card replacement.

6.7 Source Referencing

Responses can include the source used to generate the answer, such as:

- Document name
- Policy version
- Page number
- Section title
- Last updated date

6.8 Document Processing

The knowledge ingestion pipeline can process:

- PDF files
- Scanned PDF files
- Microsoft Word documents
- PowerPoint presentations
- Images
- Structured FAQ files

6.9 OCR for Scanned Documents

PaddleOCR can extract text from scanned documents and images before the content is cleaned, divided into chunks, embedded, and stored in ChromaDB.

6.10 Sensitive-Data Protection

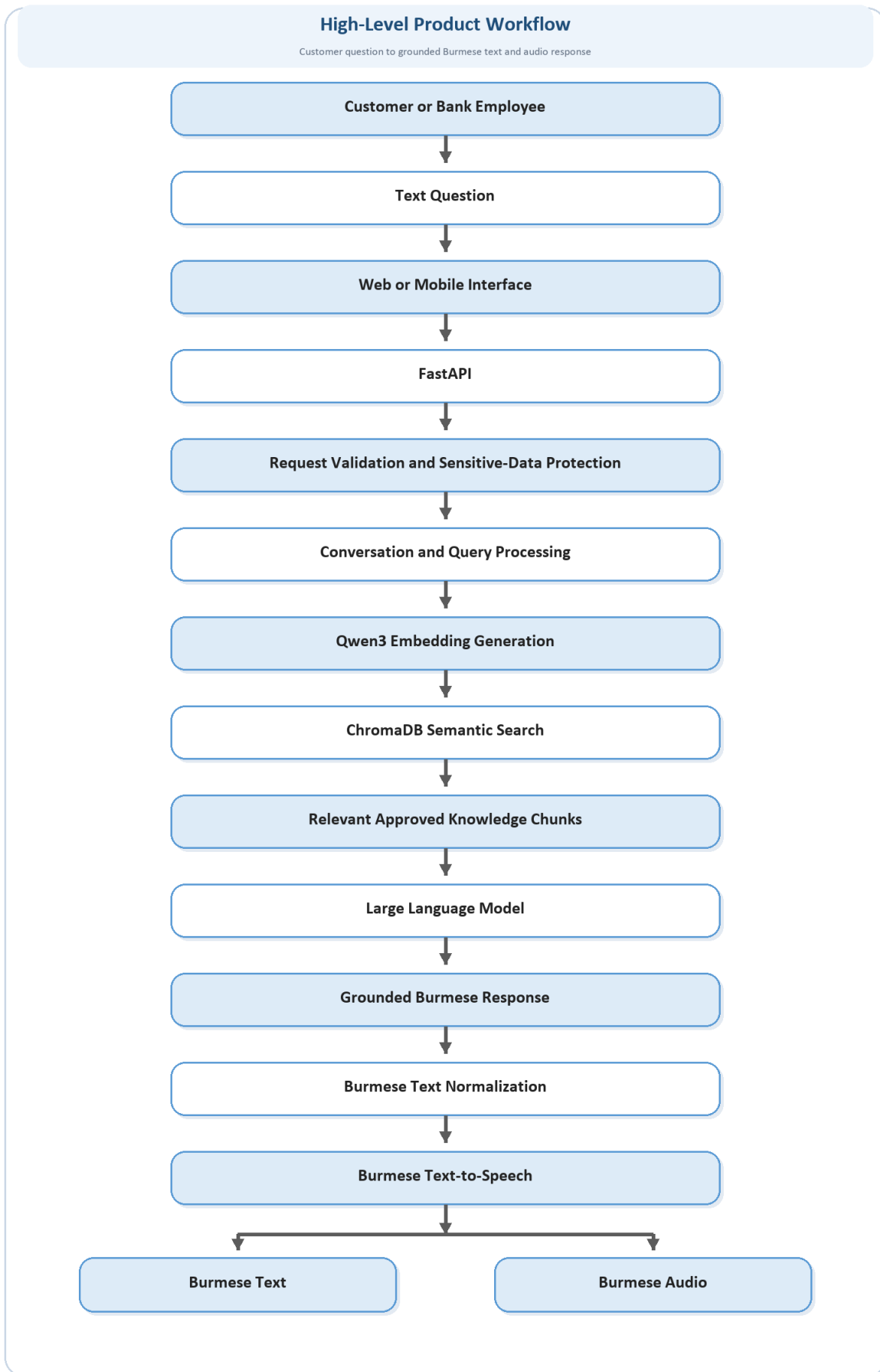
The system can detect or mask sensitive information before processing or logging it.

Examples include:

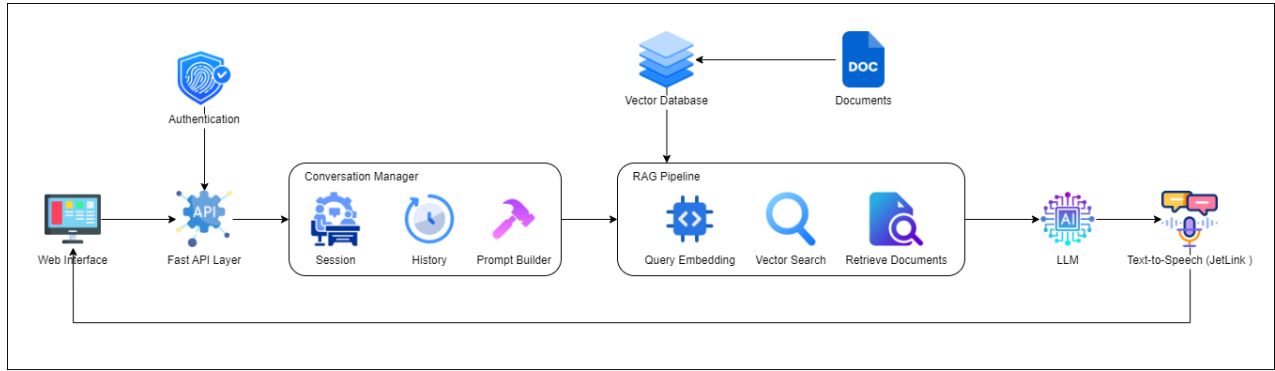
- Account numbers
- Card numbers
- NRC numbers
- Phone numbers
- PINs
- Passwords
- OTP codes

7. High-Level Product Workflow

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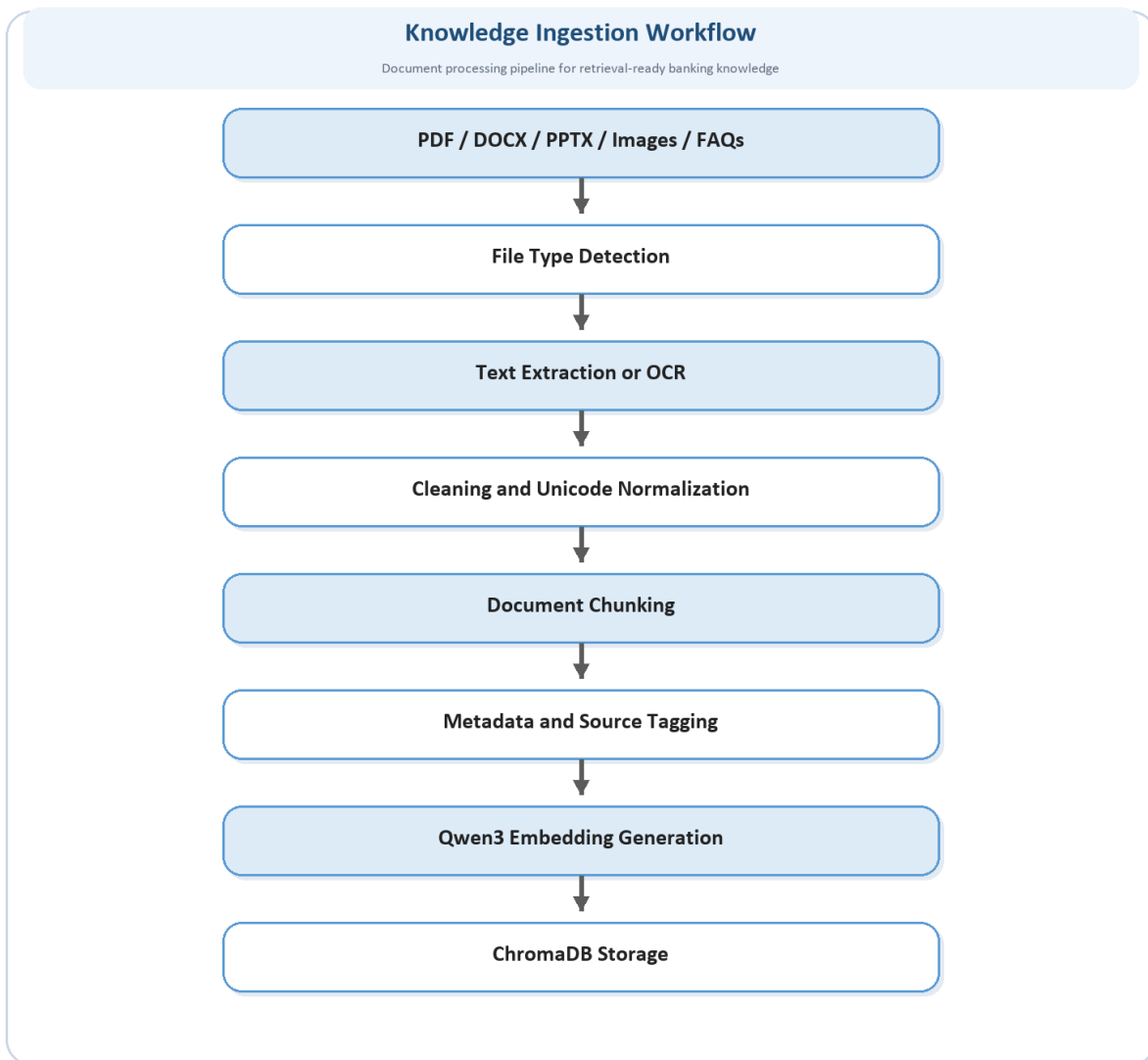


Architecture Diagram



8. Knowledge Ingestion Workflow

8. Knowledge Ingestion Workflow



Processing Components

- **PyMuPDF**: PDF extraction
- **pdfplumber**: table and layout-aware PDF extraction
- **python-docx**: Word document extraction
- **python-pptx**: PowerPoint extraction
- **PaddleOCR**: scanned document and image OCR
- **LlamaIndex**: ingestion, chunking, retrieval, and RAG orchestration
- **Qwen3 Embedding**: multilingual and Burmese-aware semantic representation
- **ChromaDB**: vector and metadata storage

9. Main System Components

9.1 User Interface

A web or mobile interface allows users to:

- Enter a question
- Read the Burmese answer
- Play or pause generated audio
- View sources
- Submit feedback
- Start a new conversation

9.2 FastAPI Backend

FastAPI manages:

- Authentication
- Request validation
- Session handling
- RAG requests
- TTS generation
- Audio delivery
- Logging
- Error handling
- Integration with future bank services

9.3 RAG Orchestration

LlamaIndex coordinates:

- Document ingestion
- Text chunking
- Embedding generation

- ChromaDB indexing
- Semantic retrieval
- Prompt construction
- Source attachment
- Context management

9.4 Qwen3 Embedding Model

The embedding layer is responsible for:

- Encoding Burmese questions
- Encoding Burmese and mixed-language documents
- Matching semantically similar content
- Supporting multilingual banking terminology
- Improving retrieval when users phrase questions differently

A suitable Qwen3 Embedding checkpoint can be selected according to deployment resources, latency requirements, and retrieval-quality testing.

9.5 ChromaDB

ChromaDB serves as the vector database for the MVP.

It stores vector embeddings together with metadata and supports:

- Similarity search
- Persistent collections
- Metadata filtering
- Local execution
- Docker deployment
- Collection separation by bank, department, product, or document type

9.6 Large Language Model

The LLM receives:

- The customer question
- Retrieved document context
- Conversation context
- Banking response rules
- Burmese-language response instructions

It produces a grounded Burmese answer and should not answer unsupported questions as confirmed facts.

9.7 Burmese Text Normalization

Before voice generation, the answer is converted into speech-friendly text.

The normalization layer handles:

- Dates
- Times
- Currency values
- Percentages
- Phone numbers
- English abbreviations
- ATM
- OTP
- PIN
- Mixed Burmese-English banking terms
- Sentence splitting and pauses

9.8 Text-to-Speech Engine

Recommended options:

- **JetlinkTTS:** self-hosted and customizable candidate
- **Azure Speech:** managed production alternative
- **Custom Burmese voice:** future bank-branded voice

The TTS service returns an audio file or audio stream to the application.

10. Proposed Technology Stack

TECHNOLOGY STACK

The proposed stack is designed for fast MVP delivery and controlled enterprise scaling, combining FastAPI, LlamaIndex, Qwen3 embeddings, ChromaDB, and Burmese TTS services.

Layer	Technology	Purpose
Frontend	React or Flutter	User-facing web or mobile experience
Backend	Python and FastAPI	API orchestration, sessions, and business logic
RAG Framework	LlamaIndex	Document ingestion, retrieval, and context assembly
PDF Processing	PyMuPDF and pdfplumber	Extract text and structure from PDFs
Word Processing	python-docx	Read Microsoft Word knowledge sources
PowerPoint Processing	python-pptx	Read PowerPoint knowledge sources

OCR	PaddleOCR	Extract text from scanned documents and images
Embedding Model	Qwen3 Embedding	Encode Burmese and multilingual semantic meaning
Vector Database	ChromaDB	Store embeddings and retrieve relevant chunks
LLM	Gemini initially, with local-model option later	Generate grounded Burmese responses
Burmese TTS	JetlinkTTS or Azure Speech	Convert response text into natural speech
Operational Database	PostgreSQL or another relational database	Store users, permissions, feedback, and audit records
Local Development	Local persistent storage	Persist local development data for the MVP
Deployment	Docker	Package and deploy the solution consistently
API Security	HTTPS, authentication, authorization, and rate limiting	Protect API access and service usage

ChromaDB stores vectors and retrieval metadata. A separate relational database may still be used for users, permissions, conversations, audit logs, feedback, and operational records.

11. Security and Banking Safety

SECURITY

The solution is positioned as a controlled information assistant with safeguards for authentication, masking, source traceability, audit logging, and human escalation.

The initial product should operate as an information and support assistant rather than a transaction-execution system.

Recommended Controls

- Authentication and role-based access
- HTTPS encryption
- Sensitive-data masking
- Secure secret management
- Audit logging

- Document access control
- Prompt-injection protection
- Retrieval filtering
- Response source tracking
- Configurable retention policies
- Human escalation for unsupported or high-risk requests

The Assistant Must Never Request

- Full PIN
- Password
- Complete card security code
- OTP
- Unnecessary full card or account details

Unsupported Questions

When reliable information cannot be retrieved, the assistant should say that it cannot confirm the answer and direct the user to an authorized bank representative.

12. Recommended MVP Scope

Included

- Text-based Burmese questions
- Approved FAQ and policy knowledge base
- Qwen3 semantic embeddings
- ChromaDB vector retrieval
- Grounded Burmese text answers
- Burmese audio output
- Source references
- Basic conversation context
- Sensitive-data masking
- Feedback collection

Excluded from the Initial MVP

- Money transfers
- Balance inquiries using live customer accounts
- PIN or password changes
- Loan approval decisions
- Card activation
- OTP processing
- Autonomous access to core banking

- High-risk financial recommendations

13. Implementation Phases

FUTURE ROADMAP

A phased roadmap moves from proof of concept to pilot deployment and production expansion without exposing high-risk banking transactions in the initial release.

Phase 1: Proof of Concept

- Prepare 50–100 approved banking FAQs.
- Build the FastAPI service.
- Generate Qwen3 embeddings.
- Store vectors in ChromaDB.
- Implement basic semantic retrieval.
- Generate Burmese answers.
- Convert answers to Burmese speech.

Phase 2: MVP

- Add document ingestion.
- Add scanned document OCR.
- Add source references.
- Add conversation context.
- Add sensitive-data masking.
- Add web interface and feedback.
- Evaluate Burmese retrieval and TTS quality.

Phase 3: Pilot Deployment

- Deploy with Docker.
- Add authentication and user roles.
- Add audit logging.
- Test with selected bank employees.
- Measure answer quality, latency, and user satisfaction.
- Improve the knowledge base and terminology normalization.

Phase 4: Production Expansion

- Integrate approved banking APIs.
- Add voice input.
- Add customer authentication.
- Add personalized but controlled responses.
- Add monitoring and analytics.

- Develop a custom bank-branded Burmese voice.

14. Evaluation Metrics

The solution should be measured using:

Retrieval

- Relevant document retrieval rate
- Top-k retrieval accuracy
- Burmese query understanding
- Mixed Burmese-English retrieval quality
- Source correctness

Response

- Factual correctness
- Groundedness
- Burmese grammar
- Clarity
- Completeness
- Unsupported-answer refusal rate

Voice

- Burmese pronunciation
- Naturalness
- Friendliness
- Gentleness
- Speaking speed
- Correct reading of numbers and banking terms
- Customer-service suitability

Business

- Reduction in average handling time
- Reduction in repetitive inquiries
- Customer satisfaction
- Agent productivity
- Knowledge-search time saved
- Escalation rate

15. Business Benefits

BUSINESS BENEFITS

The proposal improves customer accessibility and consistency while lowering repetitive support work, accelerating staff onboarding, and increasing reuse of approved banking knowledge.

Customer Benefits	Bank Benefits
Faster responses	Reduced repetitive support workload
Burmese-language accessibility	More consistent customer communication
Friendly voice guidance	Faster employee access to approved knowledge
Consistent information	Lower customer-support costs
24/7 availability for supported questions	Improved onboarding for new employees
Easier access for users who prefer audio	Better use of existing policy documents
	Scalable foundation for future AI services

16. Future Expansion

FUTURE EXPANSION

The architecture is intentionally modular, enabling voice input, employee copilots, mobile integration, multilingual support, and bank-branded AI services later.

The platform can later support:

- Burmese voice input
- Call-center agent assistance
- Automated call handling
- Mobile banking integration
- Product recommendation with strict controls
- Branch locator and appointment booking
- Fraud-awareness assistance
- Employee knowledge assistant
- Compliance knowledge search
- Multilingual customer support
- Custom bank-branded voice

17. Sales Positioning

The product should be positioned as:

A secure Burmese AI support platform that transforms approved banking knowledge into fast, consistent, and natural text-and-voice assistance.

It is more than a chatbot. It combines:

- Burmese semantic search
- Bank-controlled knowledge
- Generative AI
- Document intelligence
- Natural Burmese voice
- Modular deployment

This gives banks a practical path from a focused customer-support MVP to a broader enterprise AI assistant.

18. Conclusion

The AI-Powered Burmese Voice Support Copilot helps banks deliver faster, more accessible, and more consistent customer support. Qwen3 Embeddings provide Burmese-capable semantic retrieval, while ChromaDB offers a practical vector database for local development and MVP deployment.

By combining RAG, approved banking documents, Burmese response generation, text normalization, and Burmese TTS, the product can provide useful voice-enabled assistance without initially exposing high-risk banking transactions.

Its modular architecture allows the solution to expand into employee support, call-center assistance, mobile banking integration, and a bank-owned custom Burmese voice.